features "Micro-Ops Fusion" technology. Suppose one complex command contains more than one independent microinstruction. The decoder unit on the CPU will connect these commands to each other. Micro-Ops Fusion bundles these commands together, to be executed in a particular order. For example, an ALU operation and a Load operation will be combined to form a single Micro-Op. To the CPU, however, this encapsulation of microinstructions appears as a single command. It's only at the execution stage that the individual command threads are executed in succession, and not simultaneously.

The proverbial partner, Macro-Op Fusion, allows set of two commands can be executed as a single command. This means the four decoder units on the Core 2 Duo can in reality decode five instructions, instead of four, in a clock cycle. Such command combinations aren't always possible, but hypothetically, even if it happens only once per clock (the chances of it happening this frequently are very good) five instructions will have been executed in that cycle.

This fusion has other merits. Since the fused instructions now move along the pipeline as a single entity, buffer space is saved in case of Out of Order Execution, and the decoding bandwidth requirement is also reduced. Macro-Op Fusion increases the amount of instructions can be stuffed into the pipeline CPU processing speed, and saves power due to the fact that the CPU is able to tackle commands in queue that much faster.

Micro-Op and Macro Fusion work hand-in-glove to improve CPU load and execution efficiency.

Intelligent Power Capability—I'm The Frugal One!

The Core 2 range support Enhanced Halt State and Enhanced Intel SpeedStep. This enables huge power savings, as the CPU will throttle down when not under load. The Conroe will work from speeds of 1.6 GHz to 2.93 GHz. This down-clocking is achieved by reducing the multiplier. The Core allows for a minimum multiplier of 6; the maximum is 11, the FSB is quad pumped at 266 MHz.

Equally important—perhaps more so—is the fact that the Conroe as a whole can interactively disable any of its subsystems that aren't in use at a point in time. Intel takes pains to explain that there is no latency whatsoever involved in this On and Off switching process.

The Conroe can also dynamically switch off parts of its L2 cache that aren't in use, which was previously impossible. Traditionally in case of any cache transaction, however small in a CPU, the entire block of cache needed to be activated.

Advanced Digital Media Boost—Sounds Impressive?

Multimedia is all about SIMD (Single Instruction-Multiple Data), instruction sets. SSE (Streamed SIMD Extensions) is Intel's baby, born in 1999 and adopted first by their Pentium III processors. Applications like video and audio editing and encoding, data encryption and their ilk use a lot of SSE. Most SSE instructions today are 128-bit, which are operated on 64 bits at a time—meaning that a complete SSEI can be processed in two processor clock cycles.

The Conroe changes things with the ability to process a 128-bit SSE in a single clock cycle. SSE 4 makes its first appearance in the Conroe, consisting of eight new SSE commands—over and above the existing SSE 3 instructions.

What this should mean for home users is quicker video and audio encoding, and a richer multimedia experience—for example, DVD playback or MP3 encoding. ■

Double-Core Or Dual-Core?

Before the Core 2 Duo, Intel was basically integrating two cores on a single die—double cores. AMD 64s were the first "true dual-cores" for the desktop. The cores on an X2 processor can communicate with each other internally, via the Crossbar Switch. Pentium Ds used the external FSB to communicate with each other, which is much slower.

The Ca(t)ching Game

	INTEL				AMD	
Platform	LGA 775	Socket 478	LGA 775 / Socket 478	Socket 478	Socket 940 / 939	Socket 462
Architectural Code Name	Core (P8)	Dothan	NetBurst (P7)	NetBurst (P7)	Hammer (K8)	Barton (K7)
Processor Name	Core 2 Duo	Pentium M	Prescott	Northwood	AMD 64 / AMD 64 X2	Athlon XP
L1 Instruction Cache (KB)	32	32	16	8	64	64
L1 Data Cache (KB)	32	32	8 - 16	8 - 16	64	64
L1 Cache Latency (lesser is better)	3	3	4	2	3	3
L1 Cache Associativity	8-way	8-way	8-way	4-way	2-way	2-way
L1 Instruction TLB	128	128	128	128	32	24
L1 Data TLB	256	128	8	8	32	24
L2 Cache (KB)	4096	2048	1024 / 2048	512	1024	512
L2 Cache Associativity	16-way	8-way	8-way	8-way	16-way	16-way
L2 Cache Width (bits)	256	256	256	256	128	64
L2 TLB entries	NA	NA	64	64	512	256
Maximum memory bandwidth to CPU (GBps)	10.6	4.2	8.5	4.2	6.4	3.2

and very complex instructions are handled by a Microcode Sequencer, which is another unit placed on a CPU just to handle instructions that are too complex for the decoders. Nonetheless, in case of instructions that are complex, but not so complex as to require the Sequencer to do kick in, the three complex decoders aboard the AMD 64 processors are quite powerful.

Closing Thoughts

An architectural insight is never an easy topic to conclude. That said, the latest architectures from Intel and AMD clearly show a conscious drive towards power processing and parallelism while ensuring a declining power consumption curve—two clearly divergent paths, which means that achieving one often means compromising on the other.

Intel has already paved the way and pretty much atoned for past flaws with their brilliant Core architecture. AMD did pretty much the same thing a couple of years earlier with their much acclaimed K8—the AMD 64. The next couple of years should see the move to yet smaller fabs, as Intel plans a move to 45 nm. AMD is also looking at 65 nm for what could be their next counter-attack. Then there is the much-anticipated appearance of quad cores for the desktop. Just think about it—processing power just got a whole lot cheaper, a whole lot more powerful, and the future promises more of the same! Let's hope these promises convert to deliverances—they have thus far! ■

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